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Studies on electrophilic addition reaction of 2,3-allenoates with PhSeCl

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Abstract—β-Organoselenium substituted butenolides were prepared from 2,3-allenoates and PhSeCl in the presence of water. The yields of the products depend largely on the structures of 2,3-allenoates. The addition of water is crucial for some of this electrophilic cyclization. The reaction of simple unsubstituted methyl 2,3-butadienoate afforded methyl 4-chloro-3-phenylselanylbut-2(*Z*)-enoate in good yield and stereoselectivity.
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1. Introduction

Since the 19th century, selenium has been recognized as an essential trace element in the diet.[1] In modern synthetic organic chemistry, organoseleniums have also been considered as important synthetic intermediates.[1,2] On the other hand, butenolides are a class of compounds present in many natural and unnatural products with biological importance.[3] Butenolides are, of course, also important intermediates in organic synthesis due to the presence of the conjugated C=C bond as well as the five-membered lactone ring.[4] Thus, the preparation of organoseleno-substituted butenolides is of current interest.[5]

This group has developed some methods for the efficient synthesis of differently substituted butenolides.[6] In addition, we have also reported the electrophilic cyclization of 2,3-allenoic acids[7] with PhSeCl or PhSCl leading to the synthesis of β-organoselenium or β-organosulfur substituted butenolides in moderate yields.[8] However, 2,3-allenoic acids were usually prepared from the hydrolysis of 2,3-allenoates, in which alkynoic acids may also be formed as the byproducts.[9] In this paper, we wish to report our recent studies on the reaction of 2,3-allenoates with PhSeCl in aqueous MeCN.

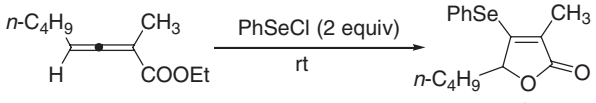
Keywords: Cyclization; Selenium; 2,3-Allenoates; Yield; Butenolide; Water effect.
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2. Results and discussion

We initiated this study with the electrophilic cyclization of ethyl 2-methyl-2,3-octadienoate **1a** with PhSeCl. However, when the reaction was carried out at room temperature in CH_3CN with 2 equiv of PhSeCl, the reaction conditions applied for the electrophilic cyclization of 2,3-allenoic acids with PhYCl (Y=S, Se),[8] the yield of **2a** was only 13% (entry 1, Table 1). Further investigation showed that water is very important in this reaction.[10] When the reaction was conducted according to the procedure described in Ref. 10, the yield of **2a** was still low and an unidentified product was formed. When a little amount of water was added to the

Table 1. Cyclization of PhSeCl with ethyl 2-methyl-2,3-octadienoate[a]



1a **2a**

Entry	CH_3CN/H_2O	Time (h)	Yield (%)[b]
1	100/0	6.5	13
2	800/1[c]	1.5	26
3	100/1[c]	1	45
4	**10/1**[c]	**1**	**100**

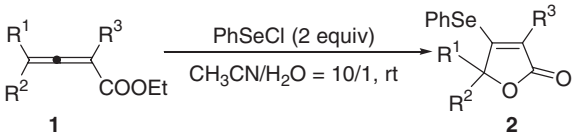
[a] The reaction was carried out using 0.2 mmol of ethyl 2-methyl-2,3-octadienoate and 2 equiv PhSeCl in 2 mL of CH_3CN.
[b] Isolated yield.
[c] H_2O was added to a solution of PhSeCl in CH_3CN, then a solution of **1a** was added immediately.

solution of PhSeCl in MeCN first, the yield of **2a** was improved (entry 2, Table 1). When $CH_3CN/H_2O=10:1$ was used, to our surprise, the yield of **2a** was quantitative (entry 4, Table 1).

With the optimized reaction conditions in hand, the scope of this reaction was explored. Some typical results are summarized in Table 2. It is obvious that differently substituted 2,4-disubstituted 2,3-allenoates can be applied to afford β-phenylselanyl butenolides in fairly good yields (77–100%) (entries 1–11, Table 2). Several types of substituents R^1 such as H, alkyl or aryl, R^2 such as H or alkyl, and R^3 such as alkyl or benzyl could be introduced into the products. The reaction of 4,4-disubstituted 2,3-allenoate **1m** afforded the product **2m** in 55% yield (entry 12, Table 2). The products were also afforded in moderate yields when 4-mono-substituted 2,3-allenoates **1n** or **1o** were treated with PhSeCl (entries 13 and 14, Table 2). In the reaction of 2-mono-substituted 2,3-allenoates **1p** or **1q**, the ratio of $MeCN/H_2O$ should be 1:1 for a higher yield of **2p** or **2q** (compare entry 15 with entry 16, Table 2). However, 2,4,4-trisubstituted 2,3-allenoates **1r** and **1s** can be applied to afford the products **2r** and **2s** in good yields even in the absence of water although the related yields are relatively lower probably due to the highly electrophilic nature of the allene moiety (entries 18–21, Table 2). Compared with results obtained with the corresponding 2,3-allenoic acids, the yields of this method are higher, in most cases, when 2,4-disubstituted 2,3-allenoate was applied. However, the reaction of 4-mono-substituted 2,3-allenoates afforded the products in relatively lower yield.

To our surprise, when we applied the current standard conditions to methyl 2,3-butadienoate **1t**, instead of

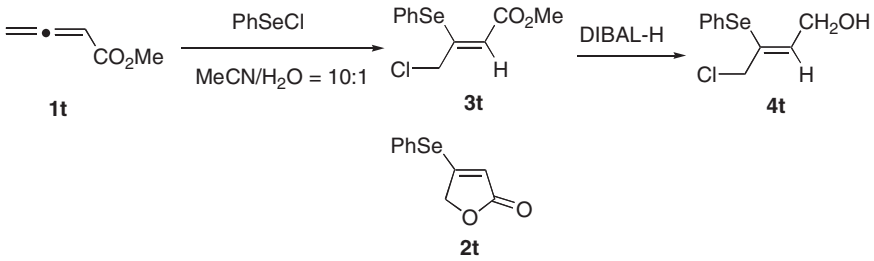
Table 2. Cyclization of PhSeCl with different ethyl 2,3-allenoate[a]



1 2

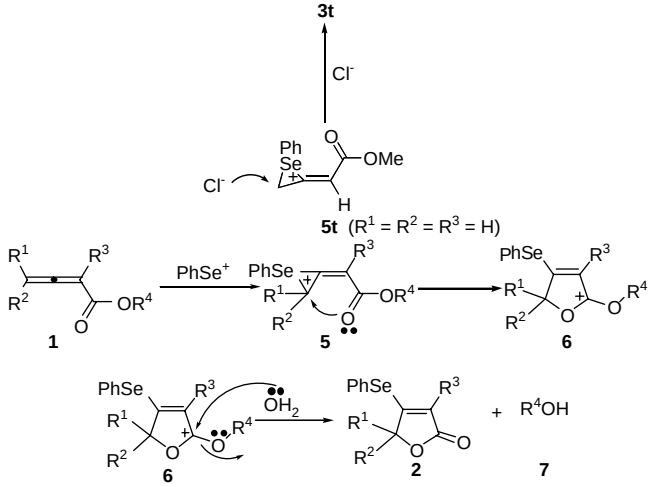
Entry	R^1	R^2	R^3	Time (h)	Yield (%)[b]
1	n-C_4H_9	H	CH_3 (**1a**)	1	100 (**2a**)
2	n-C_4H_9	H	n-C_3H_7 (**1b**)	1	92 (**2b**)
3	n-C_4H_9	H	Bn (**1c**)	1	96 (**2c**)
4	n-C_7H_{15}	H	CH_3 (**1d**)	2	94 (**2d**)
5	n-C_7H_{15}	H	n-C_3H_7 (**1e**)	1	95 (**2e**)
6	n-C_6H_{13}	H	Bn (**1f**)	3	82 (**2f**)
7	Ph	H	CH_3 (**1g**)	1	95 (**2g**)
8	Ph	H	n-C_3H_7 (**1h**)	1	99 (**2h**)
9	Ph	H	Bn (**1i**)	1	80 (**2i**)
10	α-Naphthyl	H	CH_3 (**1j**)	1	79 (**2j**)
11	α-Naphthyl	H	n-C_3H_7 (**1k**)	1	77 (**2k**)
12	CH_3	CH_3	H (**1m**)	1	55 (**2m**)
13	n-C_4H_9	H	H (**1n**)	3.5	55 (**2n**)
14	n-C_7H_{15}	H	H (**1o**)	1.5	52 (**2o**)
15	H	H	n-C_3H_7 (**1p**)	1	22 (**2p**)[c]
16[d]	H	H	n-C_3H_7 (**1p**)	1	49 (**2p**)[c]
17[d]	H	H	Bn (**1q**)	1	48 (**2q**)[c]
18	Ph	CH_3	CH_3 (**1r**)	1	100 (**2r**)
19	Ph	CH_3	CH_3 (**1r**)	1	92 (**2r**)[e]
20	$-(CH_2)_5-$		CH_3 (**1s**)	1	96 (**2s**)
21	$-(CH_2)_5-$		CH_3 (**1s**)	1	84 (**2s**)[e]

[a] The reaction was carried out using 0.2–0.3 mmol of ethyl 2,3-allenoate and 2 equiv of PhSeCl in 4–6 mL of CH_3CN.
[b] Isolated yield.
[c] Another unidentified product was also formed in this case.
[d] $CH_3CN/H_2O=1:1$.
[e] No water was added.

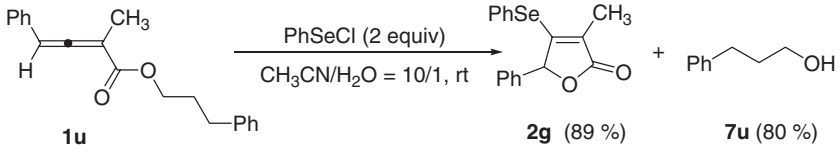


1t 3t 4t 2t

Scheme 1.



Scheme 2.



Scheme 3.

butenolide **2t**, only compound *Z*-**3t** was isolated in 81% yield. The stereoselectivity was determined by NOE study of **3t** and **4t** (Scheme 1).

A plausible mechanism for this reaction is depicted in Scheme 2. The electrophile PhSeCl reacted with the relatively electron-rich C=C bond in 2,3-allenoates **1** forming intermediate **5**, which may be followed by the intramolecular attack of the carbonyl oxygen to form the cyclic intermediate **6**. Subsequent attack of water would be followed by intramolecular proton transfer and the elimination of R^4OH to produce butenolide **2** and the alcohol **7**. In case of $R^1=R^2=R^3=H$, that is, **1t**, the related intermediate **5t** may be easily attacked by Cl^- to form **3t** due to the unsubstituted nature of the allene moiety. The stereochemical outcome is still under investigation. However, the effect of the interaction of PhSeCl with water on the reaction is still unclear.

In order to probe the mechanistic nature of this reaction, we synthesized the 3′-phenylpropyl 2-methyl-3-phenyl-2,3-butadienoate **1u**. After the reaction, we isolated the corresponding butenolide **2g** and 3-phenylpropanol **7u**[11] in 89 and 80% yields, respectively. This further supports the proposed mechanism and explains why water is important for this reaction (Scheme 3).

3. Conclusion

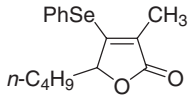
In conclusion, we have developed a facile and effective synthesis of β-organoselenium substituted butenolides from 2,3-allenoates. As a result of the easy availability of starting materials, the convenient operation and the usefulness of the products, the reaction may have potential utility in organic synthesis. The reaction of simple unsubstituted methyl 2,3-butadienoate afforded methyl 4-chloro-3-phenylselanylbut-2(*Z*)-enoate in good yield and stereoselectivity. Further studies in this area are being pursued in our laboratory.

4. Experimental

4.1. Typical procedure for the preparation of 2a

To a solution of PhSeCl (77.2 mg, 0.40 mmol) in 2 mL of MeCN was added 0.4 mL of H_2O. Then a solution of **1a** (37.1 mg, 0.20 mmol) in MeCN (2 mL) was subsequently added and the resulting mixture was stirred at room temperature for 1 h. The mixture was then evaporated directly and purified by column chromatography on silica gel (petroleum ether/ethyl acetate=10:1) to afford **2a** (64.9 mg, 100%) as a liquid.

4.1.1. 5-(*n*-Butyl)-3-methyl-4-phenylselanyl-5*H*-furan-2-one (2a).

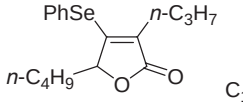


$C_{15}H_{18}O_2Se$

Compound **2a**: liquid. ^1H NMR (400 MHz, $CDCl_3$) δ 7.56 (d, $J=8.4$ Hz, 2H), 7.40–7.31 (m, 3H), 4.71–4.67 (m, 1H), 1.81 (d, $J=1.6$ Hz, 3H), 1.80–1.73 (m, 1H), 1.36–1.12 (m, 5H), 0.78 (t, $J=7.0$ Hz, 3H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 172.0, 156.0, 135.1, 129.7, 129.2, 127.5, 124.9, 82.9, 32.7, 26.1, 21.9, 13.7, 10.5; MS (70 eV, EI) *m/z* (%): 310 ($M^+(^{80}Se)$, 100), 308 ($M^+(^{78}Se)$ or $M^+(^{77}Se)+1$, 50.18) 312 ($M^+(^{82}Se)$, 19.20), 306 ($M^+(^{76}Se)$, 19.15), 307

(M^+(^{77}Se) or M^+(^{76}Se)+1, 19.31), 304 (M^+(^{74}Se), 1.87), 311 (M^+(^{80}Se)+1, 17.02); IR ν (cm^{-1}): 2957, 1755, 1627, 1401, 1280, 1087; HRMS calcd for $C_{15}H_{19}O_2^{80}Se^+$ (M^++H): 311.0545. Found: 311.0543.

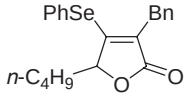
4.1.2. 5-(*n*-Butyl)-4-phenylselanyl-3-(*n*-propyl)-5*H*-furan-2-one (2b).



$C_{17}H_{22}O_2Se$

The reaction of 41.8 mg (0.20 mmol) of **1b** and 77.4 mg (0.40 mmol) of PhSeCl in 0.4 mL of H_2O and 4 mL of MeCN afforded 62.2 mg (92%) of **2b**: liquid. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.57 (d, J=7.6 Hz, 2H), 7.42–7.31 (m, 3H), 4.67–4.64 (m, 1H), 2.40–2.30 (m, 1H), 2.29–2.19 (m, 1H), 1.80–1.71 (m, 1H), 1.60–1.49 (m, 2H), 1.38–1.06 (m, 5H), 0.91 (t, J=7.4 Hz, 3H), 0.77 (t, J=7.2 Hz, 3H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 171.6, 156.5, 135.3, 131.6, 129.8, 129.3, 125.0, 82.5, 32.8, 27.3, 25.9, 22.0, 21.0, 13.8, 13.7; MS (70 eV, EI) *m/z* (%): 338 (M^+(^{80}Se), 39.97), 336 (M^+(^{78}Se) or M^+(^{77}Se)+1, 33.74), 340 (M^+(^{82}Se), 24.20), 334 (M^+(^{76}Se), 7.45), 335 (M^+(^{77}Se) or M^+(^{76}Se)+1, 25.43), 333 (M^+(^{74}Se)+1, 2.65), 337 (M^+(^{78}Se)+1, 54.80), 341 (M^+(^{82}Se)+1, 20.66), 339 (M^+(^{80}Se)+1, 100); IR ν (cm^{-1}): 2959, 2931, 1754, 1621; HRMS calcd for $C_{17}H_{22}O_2^{80}Se$: 338.0791. Found: 338.0797.

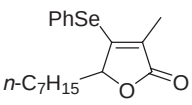
4.1.3. 3-Benzyl-5-(*n*-butyl)-4-phenylselanyl-5*H*-furan-2-one (2c).



$C_{21}H_{22}O_2Se$

The reaction of 52.6 mg (0.20 mmol) of **1c** and 79.0 mg (0.41 mmol) of PhSeCl in 0.4 mL of H_2O and 4 mL of MeCN afforded 75.8 mg (96%) of **2c**: liquid. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.52 (d, J=8.0 Hz, 2H), 7.40–7.18 (m, 8H), 4.66 (dd, J_1=7.6 Hz, J_2=3.2 Hz, 1H), 3.76 (d, J=14.4 Hz, 1H), 3.57 (d, J=14.4 Hz, 1H), 1.76–1.64 (m, 1H), 1.28–1.00 (m, 5H), 0.73 (t, J=7.0 Hz, 3H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 171.4, 157.9, 137.5, 135.4, 130.2, 129.8, 129.5, 128.7, 128.5, 126.6, 124.8, 82.7, 32.8, 31.4, 26.0, 21.9, 13.6; MS (70 eV, EI) *m/z* (%): 386 (M^+(^{80}Se), 39.73), 384 (M^+(^{78}Se) or M^+(^{77}Se)+1, 34.01), 388 (M^+(^{82}Se), 26.58), 382 (M^+(^{76}Se), 8.28), 383 (M^+(^{77}Se) or M^+(^{76}Se)+1, 26.08), 387 (M^+(^{80}Se)+1, 98.17), 385 (M^+(^{78}Se)+1, 57.04), 389 (M^+(^{82}Se)+1, 19.54), 381 (M^+(^{74}Se)+1, 3.55), 115 (100); IR ν (cm^{-1}): 2956, 2928, 1751, 1618, 1063; HRMS calcd for $C_{21}H_{22}O_2^{80}Se$: 386.0793. Found: 386.0808.

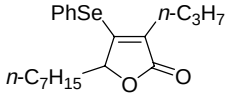
4.1.4. 5-(*n*-Heptyl)-3-methyl-4-phenylselanyl-5*H*-furan-2-one (2d).



$C_{18}H_{24}O_2Se$

The reaction of 43.7 mg (0.20 mmol) of **1d** and 75.2 mg (0.40 mmol) of PhSeCl in 0.4 mL of H_2O and 4 mL of MeCN afforded 64.7 mg (94%) of **2d**: liquid. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.56 (d, J=8.0 Hz, 2H), 7.42–7.31 (m, 3H), 4.72–4.70 (m, 1H), 1.82 (d, J=2.0 Hz, 3H), 1.79–1.69 (m, 1H), 1.36–1.00 (m, 11H), 0.85 (t, J=7.2 Hz, 3H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 172.0, 156.1, 135.1, 129.7, 129.2, 127.5, 124.9, 82.9, 33.1, 31.6, 28.8, 24.0, 22.5, 14.0, 10.5; MS (70 eV, EI) *m/z* (%): 352 (M^+(^{80}Se), 45.68), 350 (M^+(^{78}Se) or M^+(^{77}Se)+1, 36.25), 354 (M^+(^{82}Se), 25.58), 348 (M^+(^{76}Se), 8.67), 349 (M^+(^{77}Se) or M^+(^{76}Se)+1, 26.78), 353 (M^+(^{80}Se)+1, 98.04), 351 (M^+(^{78}Se)+1, 53.98), 355 (M^+(^{82}Se)+1, 18.24), 347 (M^+(^{74}Se)+1, 3.10), 41 (100); IR ν (cm^{-1}): 2926, 2856, 1755, 1628, 1281, 1091; HRMS calcd for $C_{18}H_{24}O_2^{80}Se$: 352.0948. Found: 352.0957.

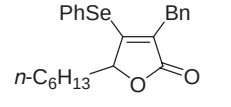
4.1.5. 5-(*n*-Heptyl)-4-phenylselanyl-3-(*n*-propyl)-5*H*-furan-2-one (2e).



$C_{20}H_{28}O_2Se$

The reaction of 51.7 mg (0.21 mmol) of **1e** and 76.1 mg (0.40 mmol) of PhSeCl in 0.4 mL of H_2O and 4 mL of MeCN afforded 73.7 mg (95%) of **2e**: liquid. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.57 (d, J=8.0 Hz, 2H), 7.40–7.31 (m, 3H), 4.66–4.64 (m, 1H), 2.37–2.32 (m, 1H), 2.26–2.21 (m, 1H), 1.74–1.71 (m, 1H), 1.60–1.48 (m, 2H), 1.40–1.00 (m, 11H), 0.90 (t, J=7.2 Hz, 3H), 0.85 (t, J=7.0 Hz, 3H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 171.6, 156.5, 135.3, 131.6, 129.7, 129.3, 125.0, 82.5, 33.1, 31.6, 28.80, 28.77, 27.3, 23.8, 22.5, 21.0, 14.0, 13.7; MS (70 eV, EI) *m/z* (%): 380 (M^+(^{80}Se), 24.79), 378 (M^+(^{78}Se) or M^+(^{77}Se)+1, 20.09), 382 (M^+(^{82}Se), 14.94), 376 (M^+(^{76}Se), 4.93), 377 (M^+(^{77}Se) or M^+(^{76}Se)+1, 16.41), 381 (M^+(^{80}Se)+1, 57.51), 379 (M^+(^{78}Se)+1, 32.78), 383 (M^+(^{82}Se)+1, 11.69), 375 (M^+(^{74}Se)+1, 2.09), 41 (100); IR ν (cm^{-1}): 2958, 2928, 1755, 1622, 1459, 1065; HRMS calcd for $C_{20}H_{28}O_2^{80}Se$: 380.1262. Found: 380.1284.

4.1.6. 3-Benzyl-5-(*n*-hexyl)-4-phenylselanyl-5*H*-furan-2-one (2f).

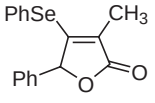


$C_{23}H_{26}O_2Se$

The reaction of 56.8 mg (0.20 mmol) of **1f** and 76.2 mg (0.40 mmol) of PhSeCl in 0.4 mL of H_2O and 4 mL of MeCN afforded 67.5 mg (82%) of **2f**: liquid. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.58 (d, J=8.0 Hz, 2H), 7.46–7.22 (m, 8H), 4.72 (dd, J_1=7.6 Hz, J_2=2.8 Hz, 1H), 3.81 (d, J=14.8 Hz, 1H), 3.62 (d, J=14.8 Hz, 1H), 1.74–1.71 (m, 1H), 1.39–1.02 (m, 9H), 0.86 (t, J=7.2 Hz, 3H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 171.4, 157.9, 137.5, 135.4, 130.2, 129.8, 129.5, 128.7, 128.5, 126.5, 124.8, 82.7, 33.1, 31.4, 31.3, 28.4, 23.7, 22.3, 13.9; MS (70 eV, EI) *m/z* (%): 414 (M^+(^{80}Se), 35.39), 412 (M^+(^{78}Se) or M^+(^{77}Se)+1, 31.77), 416 (M^+(^{82}Se), 27.40), 410 (M^+(^{76}Se), 6.10), 411

($M^+(^{77}Se)$ or $M^+(^{76}Se)+1$, 24.19), 415 ($M^+(^{80}Se)+1$, 98.21), 413 ($M^+(^{78}Se)+1$, 56.87), 417 ($M^+(^{82}Se)+1$, 19.99), 409 ($M^+(^{74}Se)+1$, 3.50), 43 (100); IR ν (cm^{-1}): 2927, 2857, 1751, 1618, 1578, 1454, 1439, 1082; HRMS calcd for $C_{23}H_{26}O_2^{80}Se$: 414.1107. Found: 414.1120.

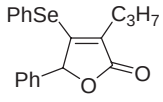
4.1.7. 3-Methyl-5-phenyl-4-phenylselanyl-5*H*-furan-2-one (2g).



$C_{17}H_{14}O_2Se$

The reaction of 39.9 mg (0.20 mmol) of **1g** and 75.9 mg (0.40 mmol) of PhSeCl in 0.4 mL of H_2O and 4 mL of MeCN afforded 61.8 mg (95%) of **2g**: liquid. ^1H NMR (300 MHz, $CDCl_3$) δ 7.38–7.15 (m, 8H), 6.86 (d, $J=8.4$ Hz, 2H), 5.56 (q, $J=1.6$ Hz, 1H), 1.92 (d, $J=1.6$ Hz, 3H); ^{13}C NMR (75 MHz, $CDCl_3$) δ 171.9, 156.4, 135.9, 134.2, 129.4, 129.3, 129.2, 128.5, 127.5, 126.6, 123.8, 84.7, 10.5; MS (70 eV, EI) *m/z* (%): 330 ($M^+(^{80}Se)$, 51.07), 328 ($M^+(^{78}Se)$ or $M^+(^{77}Se)+1$, 25.82), 332 ($M^+(^{82}Se)$, 10.77), 326 ($M^+(^{76}Se)$, 9.96), 327 ($M^+(^{77}Se)$ or $M^+(^{76}Se)+1$, 11.40), 331 ($M^+(^{80}Se)+1$, 8.42), 329 ($M^+(^{78}Se)+1$, 23.14), 333 ($M^+(^{82}Se)+1$, 8.07), 105 (100); IR ν (cm^{-1}): 3031, 2918, 1759, 1630, 1440, 1302, 1284, 1107, 1087. Anal. Calcd for $C_{17}H_{14}O_2Se$: C 62.03, H 4.29. Found: C 62.01, H 4.27.

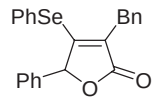
4.1.8. 5-Phenyl-4-phenylselanyl-3-(*n*-propyl)-5*H*-furan-2-one (2h).



$C_{19}H_{18}O_2Se$

The reaction of 45.6 mg (0.20 mmol) of **1h** and 79.2 mg (0.41 mmol) of PhSeCl in 0.4 mL of H_2O and 4 mL of MeCN afforded 70.0 mg (99%) of **2h**: liquid. ^1H NMR (400 MHz, $CDCl_3$) δ 7.29–7.08 (m, 8H), 6.74 (d, $J=6.8$ Hz, 2H), 5.44 (s, 1H), 2.42–2.34 (m, 1H), 2.29–2.20 (m, 1H), 1.65–1.54 (m, 2H), 0.93 (t, $J=7.4$ Hz, 3H); ^{13}C NMR (75 MHz, $CDCl_3$) δ 171.2, 156.6, 135.9, 134.2, 130.5, 129.2, 129.1, 129.0, 128.3, 127.3, 123.6, 84.2, 27.2, 20.7, 13.8; MS (70 eV, EI) *m/z* (%): 358 ($M^+(^{80}Se)$, 69.96), 356 ($M^+(^{78}Se)$ or $M^+(^{77}Se)+1$, 38.30), 360 ($M^+(^{82}Se)$, 21.28), 354 ($M^+(^{76}Se)$, 14.63), 355 ($M^+(^{77}Se)$ or $M^+(^{76}Se)+1$, 17.96), 352 ($M^+(^{74}Se)$, 1.54), 359 ($M^+(^{80}Se)+1$, 63.26), 357 ($M^+(^{78}Se)+1$, 33.41), 361 ($M^+(^{82}Se)+1$, 13.45), 353 ($M^+(^{74}Se)+1$, 1.80), 105 (100); IR ν (cm^{-1}): 2960, 2930, 2871, 1755, 1618, 1456, 1295, 1116. Anal. Calcd for $C_{19}H_{18}O_2Se$: C 63.88, H 5.08. Found: C 63.81, H 5.10.

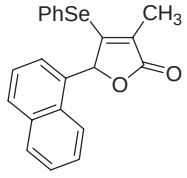
4.1.9. 3-Benzyl-5-phenyl-4-phenylselanyl-5*H*-furan-2-one (2i).



$C_{23}H_{18}O_2Se$

The reaction of 52.1 mg (0.19 mmol) of **1i** and 79.3 mg (0.41 mmol) of PhSeCl in 0.4 mL of H_2O and 4 mL of MeCN afforded 60.5 mg (80%) of **2i**: liquid. ^1H NMR (400 MHz, $CDCl_3$) δ 7.48–7.38 (m, 5H), 7.37–7.32 (m, 2H), 7.29–7.21 (m, 6H), 6.81 (d, $J=7.2$ Hz, 2H), 5.60 (s, 1H), 3.94 (d, $J=14.8$ Hz, 1H), 3.77 (d, $J=14.8$ Hz, 1H); ^{13}C NMR (75 MHz, $CDCl_3$) δ 171.2, 158.0, 137.3, 136.2, 134.2, 129.5, 129.4, 129.2, 128.8, 128.7, 128.5, 127.6, 126.7, 123.6, 84.5, 31.5; MS (70 eV, EI) *m/z* (%): 406 ($M^+(^{80}Se)$, 40.50), 404 ($M^+(^{78}Se)$ or $M^+(^{77}Se)+1$, 24.72), 408 ($M^+(^{82}Se)$, 15.13), 402 ($M^+(^{76}Se)$, 8.71), 403 ($M^+(^{77}Se)$ or $M^+(^{76}Se)+1$, 11.53), 400 ($M^+(^{74}Se)$, 1.18), 407 ($M^+(^{80}Se)+1$, 37.57), 405 ($M^+(^{78}Se)+1$, 21.48), 409 ($M^+(^{82}Se)+1$, 8.34), 401 ($M^+(^{74}Se)+1$, 0.23), 203 (100); IR ν (cm^{-1}): 3030, 2907, 1740, 1607, 1455, 1175, 1082, 1064. Anal. Calcd for $C_{23}H_{18}O_2Se$: C 68.16, H 4.48. Found: C 68.39, H 4.66.

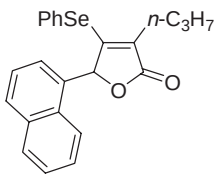
4.1.10. 3-Methyl-5-(1′-naphthyl)-4-phenylselanyl-5*H*-furan-2-one (2j).



$C_{21}H_{16}O_2Se$

The reaction of 53 mg (0.21 mmol) of **1j** and 77 mg (0.40 mmol) of PhSeCl in 0.4 mL of H_2O and 4 mL of MeCN afforded 63 mg (79%) of **2j**: solid. Mp 149–150 °C (petroleum ether/ethyl acetate). ^1H NMR (300 MHz, $CDCl_3$) δ 7.83 (t, $J=7.2$ Hz, 2H), 7.67 (d, $J=8.4$ Hz, 1H), 7.48–7.37 (m, 3H), 7.28 (d, $J=7.5$ Hz, 1H), 7.12 (d, $J=6.6$ Hz, 2H), 7.04 (t, $J=7.4$ Hz, 1H), 6.91 (t, $J=7.2$ Hz, 2H), 6.38 (q, $J=1.8$ Hz, 1H), 2.00 (d, $J=1.8$ Hz, 3H); ^{13}C NMR (75 MHz, $CDCl_3$) δ 171.9, 156.8, 135.7, 133.5, 131.6, 130.04, 130.02, 129.0, 128.9, 128.5, 127.2, 126.4, 125.9, 125.7, 125.0, 123.3, 122.1, 80.5, 10.6; MS (70 eV, EI) *m/z* (%): 380 ($M^+(^{80}Se)$, 81.24), 378 ($M^+(^{78}Se)$ or $M^+(^{77}Se)+1$, 44.30), 382 ($M^+(^{82}Se)$, 17.34), 376 ($M^+(^{76}Se)$, 15.68), 377 ($M^+(^{77}Se)$ or $M^+(^{76}Se)+1$, 17.69), 374 ($M^+(^{74}Se)$, 1.56), 381 ($M^+(^{80}Se)+1$, 19.74), 379 ($M^+(^{78}Se)+1$, 11.13), 383 ($M^+(^{82}Se)+1$, 3.48), 45 (100); IR ν (cm^{-1}): 1737, 1623, 1280, 1091. Anal. Calcd for $C_{21}H_{16}O_2Se$: C 66.50, H 4.25. Found: C 66.63, H 4.30.

4.1.11. 5-(1′-Naphthyl)-4-phenylselanyl-3-(*n*-propyl)-5*H*-furan-2-one (2k).

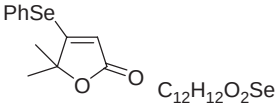


$C_{23}H_{20}O_2Se$

The reaction of 55.1 mg (0.20 mmol) of **1k** and 77.8 mg (0.41 mmol) of PhSeCl in 0.4 mL of H_2O and 4 mL of MeCN afforded 61.4 mg (77%) of **2k**: solid. Mp 105–107 °C (hexane/ethyl acetate). ^1H NMR (400 MHz, $CDCl_3$) δ 7.73 (t, $J=8.8$ Hz, 2H), 7.53 (d, $J=8.8$ Hz, 1H), 7.34

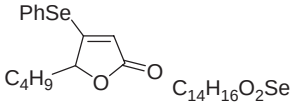
(q, J=7.6 Hz, 2H), 7.26 (t, J=7.8 Hz, 1H), 7.19 (d, J=6.8 Hz, 1H), 7.01 (d, J=7.2 Hz, 2H), 6.93 (t, J=7.4 Hz, 1H), 6.79 (t, J=7.4 Hz, 2H), 6.26 (s, 1H), 2.50–2.41 (m, 1H), 2.40–2.30 (m, 1H), 1.70–1.63 (m, 2H), 1.00 (t, J=7.4 Hz, 3H); ^{13}C NMR (75 MHz, $CDCl_3$) δ 171.5, 157.1, 136.0, 133.5, 131.7, 131.4, 130.2, 130.0, 129.05, 129.00, 128.5, 126.4, 125.9, 125.7, 125.1, 123.3, 122.1, 80.3, 27.6, 21.0, 14.1; MS (70 eV, EI) *m/z* (%): 408 (M^+(^{80}Se), 79.87), 406 (M^+(^{78}Se) or M^+(^{77}Se)+1, 41.82), 410 (M^+(^{82}Se), 16.71), 404 (M^+(^{76}Se), 15.31), 405 (M^+(^{77}Se) or M^+(^{76}Se)+1, 15.91), 402 (M^+(^{74}Se), 1.98), 409 (M^+(^{80}Se)+1, 20.27), 407 (M^+(^{78}Se)+1, 10.53), 411 (M^+(^{82}Se)+1, 3.93), 127 (100); IR ν (cm^{-1}): 2960, 2930, 2871, 1733, 1621, 1438, 1319, 1067. Anal. Calcd for $C_{23}H_{20}O_2Se$: C 67.82, H 4.95. Found: C 67.71, H 4.86.

4.1.12. 5,5-Dimethyl-4-phenylselanyl-5*H*-furan-2-one (2m).[12]



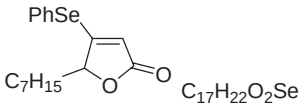
The reaction of 42.4 mg (0.30 mmol) of **1m** and 116.8 mg (0.61 mmol) of PhSeCl in 0.6 mL of H_2O and 6 mL of MeCN afforded 44.6 mg (55%) of **2m**: liquid. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.62 (d, J=6.4 Hz, 2H), 7.49–7.38 (m, 3H), 5.29 (s, 1H), 1.56 (s, 6H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 176.0, 169.9, 136.0, 130.2, 130.1, 124.6, 114.5, 87.5, 26.9; MS (EI) *m/z* (%): 268 (M^+(^{80}Se), 100), 266 (M^+(^{78}Se) or M^+(^{77}Se)+1, 52.38), 270 (M^+(^{82}Se), 23.81), 264 (M^+(^{76}Se), 23.80), 265 (M^+(^{77}Se) or M^+(^{76}Se)+1, 22.62), 262 (M^+(^{74}Se), 3.57), 269 (M^+(^{80}Se)+1, 19.05), 267 (M^+(^{78}Se)+1, 9.52), 271 (M^+(^{82}Se)+1, 4.76); IR ν (cm^{-1}): 2980, 1747, 1573, 1239, 1117.

4.1.13. 5-(*n*-Butyl)-4-phenylselanyl-5*H*-furan-2-one (2n).



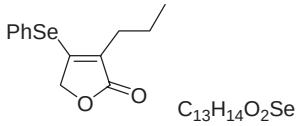
The reaction of 36.9 mg (0.22 mmol) of **1n** and 77.9 mg (0.41 mmol) of PhSeCl in 0.4 mL of H_2O and 4 mL of MeCN afforded 35.8 mg (55%) of **2n**: liquid. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.63 (d, J=6.8 Hz, 2H), 7.50–7.39 (m, 3H), 5.47 (d, J=0.8 Hz,1H), 5.06–5.01 (m, 1H), 1.96–1.88 (m, 1H), 1.68–1.57 (m, 1H), 1.53–1.28 (m, 4H), 0.92 (t, J=7.0 Hz, 3H); ^{13}C NMR (75 MHz, $CDCl_3$) δ 171.1, 169.7, 135.9, 130.2, 130.1, 124.5, 115.9, 84.2, 33.7, 26.0, 22.3, 13.8; MS (70 eV, EI) *m/z* (%): 296 (M^+(^{80}Se), 73.33), 294 (M^+(^{78}Se) or M^+(^{77}Se)+1, 38.51), 298 (M^+(^{82}Se), 1.60), 293 (M^+(^{77}Se) or M^+(^{76}Se)+1, 15.54), 297 (M^+(^{80}Se)+1, 10.86), 295 (M^+(^{78}Se)+1, 5.56), 299 (M^+(^{82}Se)+1, 4.36), 239 (100); IR ν (cm^{-1}): 2957, 2931, 1748, 1571, 1439, 1246, 1159. Anal. Calcd for $C_{14}H_{16}O_2Se$: C 56.97, H 5.46. Found: C 57.21, H 5.70.

4.1.14. 5-(*n*-Heptyl)-4-phenylselanyl-5*H*-furan-2-one (2o).



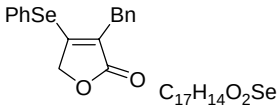
The reaction of 43.6 mg (0.21 mmol) of **1o** and 77.2 mg (0.40 mmol) of PhSeCl in 0.4 mL of H_2O and 4 mL of MeCN afforded 36.7 mg (52%) of **2o**: liquid. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.62 (d, J=8.4 Hz, 2H), 7.49–7.39 (m, 3H), 5.47 (d, J=1.6 Hz, 1H), 5.06–5.01 (m, 1H), 1.96–1.86 (m, 1H), 1.68–1.56 (m, 1H), 1.53–1.28 (m, 10H), 0.89 (t, J=6.8 Hz, 3H); ^{13}C NMR (75 MHz, $CDCl_3$) δ 171.2, 169.8, 135.9, 130.2, 130.0, 124.4, 115.9, 84.2, 33.9, 31.6, 29.1, 28.9, 23.9, 22.5, 14.0; MS (70 eV, EI) *m/z* (%): 338 (M^+(^{80}Se), 56.96), 336 (M^+(^{78}Se) or M^+(^{77}Se)+1, 32.34), 340 (M^+(^{82}Se), 27.85), 335 (M^+(^{77}Se) or M^+(^{76}Se)+1, 16.41), 334 (M^+(^{76}Se), 11.04), 339 (M^+(^{80}Se)+1, 53.15), 337 (M^+(^{78}Se)+1, 30.09), 341 (M^+(^{82}Se)+1, 4.74), 41 (100); IR ν (cm^{-1}): 2927, 2856, 1751, 1572, 1440, 1245, 1158. Anal. Calcd for $C_{17}H_{22}O_2Se$: C 60.54, H 6.57. Found: C 60.47, H 6.80.

4.1.15. 4-Phenylselanyl-3-(*n*-propyl)-5*H*-furan-2-one (2p).



The reaction of 47.6 mg (0.31 mmol) of **1p** and 117.1 mg (0.61 mmol) of PhSeCl in 0.6 mL of H_2O and 6 mL of MeCN afforded 19.2 mg (22%) of **2p**: liquid. The reaction of 47.3 mg (0.31 mmol) of **1p** and 111.5 mg (0.58 mmol) of PhSeCl in 2 mL of H_2O and 2 mL of MeCN afforded 42.3 mg (49%) of **2p**. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.64 (d, J=7.2 Hz, 2H), 7.46 (t, J=7.2 Hz, 1H), 7.38 (t, J=7.4 Hz, 2H), 4.38 (s, 2H), 2.31 (t, J=7.4 Hz, 2H), 1.66–1.55 (m, 2H), 0.96 (t, J=7.4 Hz, 3H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 172.4, 153.7, 136.3, 129.93, 129.91, 128.8, 123.0, 72.1, 27.3, 20.7, 13.9; MS (EI) *m/z* (%): 282 (M^+(^{80}Se), 100), 280 (M^+(^{78}Se) or M^+(^{77}Se)+1, 53.33), 284 (M^+(^{82}Se), 23.11), 278 (M^+(^{76}Se), 21.33), 279 (M^+(^{77}Se) or M^+(^{76}Se)+1, 20.44), 276 (M^+(^{74}Se), 1.11), 283 (M^+(^{80}Se)+1, 17.78), 281 (M^+(^{78}Se)+1, 9.33), 285 (M^+(^{82}Se)+1, 6.22), 277 (M^+(^{74}Se)+1, 0.44); IR ν (cm^{-1}): 2960, 1755, 1619, 1436, 1058, 1019; HRMS calcd for $C_{13}H_{14}O_2^{80}SeNa^+$ (M^++Na): 305.0061. Found: 305.0047.

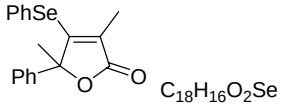
4.1.16. 3-Benzyl-4-phenylselanyl-5*H*-furan-2-one (2q).



The reaction of 59.8 mg (0.30 mmol) of **1q** and 115.3 mg (0.60 mmol) of PhSeCl in 2 mL of H_2O and 2 mL of MeCN afforded 46.8 mg (48%) of **2q**: liquid. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.52 (d, J=8.0 Hz, 2H), 7.37 (t, J=7.6 Hz, 1H), 7.32–7.21 (m, 6H), 7.20–7.15 (m, 1H), 4.28 (s, 2H), 3.60

(s, 2H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 172.2, 155.1, 137.2, 136.3, 130.0, 129.9, 129.0, 128.6, 127.4, 126.8, 122.9, 72.2, 31.4; MS (EI) *m/z* (%): 330 (M^+(^{80}Se), 23.75), 328 (M^+(^{78}Se) or M^+(^{77}Se)+1, 11.85), 332 (M^+(^{82}Se), 4.65), 326 (M^+(^{76}Se), 4.53), 327 (M^+(^{77}Se) or M^+(^{76}Se)+1, 4.58), 324 (M^+(^{74}Se), 0.41), 331 (M^+(^{80}Se)+1, 4.53), 329 (M^+(^{78}Se)+1, 2.53), 333 (M^+(^{82}Se)+1, 0.89), 129 (100); IR ν (cm^{-1}): 3059, 1747, 1618, 1454, 1439, 1341, 1172, 1055, 1023; HRMS calcd for $C_{17}H_{14}O_2^{80}SeNa^+$ (M^++Na): 353.0063. Found: 353.0077.

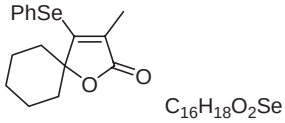
4.1.17. 3,5-Dimethyl-5-phenyl-4-phenylselanyl-5*H*-furan-2-one (2r).



$C_{18}H_{16}O_2Se$

The reaction of 40.8 mg (0.19 mmol) of **1r** and 76.9 mg (0.40 mmol) of PhSeCl in 0.4 mL of H_2O and 4 mL of MeCN afforded 64.5 mg (100%) of **2r**. The reaction of 42.9 mg (0.20 mmol) of **1r** and 78.3 mg (0.41 mmol) of PhSeCl in 4 mL of MeCN afforded 62.6 mg (92%) of **2r**: liquid. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.24–7.17 (m, 8H), 7.11 (t, J=7.6 Hz, 2H), 1.85 (s, 3H), 1.54 (s, 3H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 172.1, 160.9, 138.1, 134.2, 129.4, 128.6, 128.5, 128.3, 125.7, 125.4, 89.1, 24.6, 10.5; MS (EI) *m/z* (%): 344 (M^+(^{80}Se), 62.65), 342 (M^+(^{78}Se) or M^+(^{77}Se)+1, 32.53), 346 (M^+(^{82}Se), 14.45), 340 (M^+(^{76}Se), 13.25), 341 (M^+(^{77}Se) or M^+(^{76}Se)+1, 13.20), 338 (M^+(^{74}Se), 1.20), 345 (M^+(^{80}Se)+1, 14.50), 343 (M^+(^{78}Se)+1, 7.23), 347 (M^+(^{82}Se)+1, 2.41), 187 (100); IR ν (cm^{-1}): 3063, 1751, 1615, 1481, 1440, 1053, 1028; HRMS calcd for $C_{18}H_{16}O_2Se^{80}Na^+$ (M^++Na): 367.0220. Found: 367.0231.

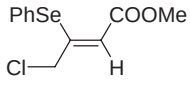
4.1.18. 3-Methyl-4-phenylselanyl-1-oxaspiro[4.5]dec-3-en-2-one (2s).



$C_{16}H_{18}O_2Se$

The reaction of 38.0 mg (0.20 mmol) of **1s** and 76.5 mg (0.40 mmol) of PhSeCl in 0.4 mL of H_2O and 4 mL of MeCN afforded 60.2 mg (96%) of **2s**. The reaction of 39.4 mg (0.20 mmol) of **1s** and 76.1 mg (0.40 mmol) of PhSeCl in 4 mL of MeCN afforded 54.6 mg (84%) of **2s**: solid. Mp 63–65 °C (hexane/ethyl acetate). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.41 (d, J=8.0 Hz, 2H), 7.30–7.20 (m, 3H), 1.79–1.59 (m, 7H), 1.47 (s, 3H), 1.46 (d, J=9.6 Hz, 2H), 1.15–1.06 (m, 1H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 171.9, 160.3, 133.8, 129.5, 128.5, 126.5, 89.1, 34.8, 24.3, 22.0, 10.5; MS (70 eV, EI) *m/z* (%): 322 (M^+(^{80}Se), 72.04), 320 (M^+(^{78}Se) or M^+(^{77}Se)+1, 37.71), 324 (M^+(^{82}Se), 14.70), 319 (M^+(^{77}Se) or M^+(^{76}Se)+1, 16.60), 318 (M^+(^{76}Se), 17.66), 316 (M^+(^{74}Se), 2.67), 323 (M^+(^{80}Se)+1, 14.61), 325 (M^+(^{82}Se)+1, 3.45), 165 (100); IR ν (cm^{-1}): 2935, 2858, 1751, 1635, 1575, 1475, 1449, 1439, 1291, 1283, 1199, 1097. Anal. Calcd for $C_{16}H_{18}O_2Se$: C 59.83, H 5.65. Found: C 60.06, H 5.73.

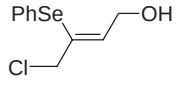
4.1.19. Methyl 4-chloro-3-phenylselanylbut-2(*Z*)-enoate (3t).



$C_{11}H_{11}ClO_2Se$

The reaction of 101.1 mg (1.03 mmol) of **3t** and 384.1 mg (2.01 mmol) of PhSeCl in 2 mL of H_2O and 20 mL of MeCN afforded 242.0 mg (81%) of **3t**: liquid. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.71 (d, J=8.0 Hz, 2H), 7.45 (t, J=7.6 Hz, 1H), 7.36 (t, J=7.2 Hz, 2H), 6.60 (t, J=1.1 Hz, 1H), 3.95 (d, J=1.1 Hz, 2H), 3.79 (s, 3H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 166.9, 154.1, 137.6, 129.7, 129.3, 125.3, 116.0, 51.6, 46.7; MS (EI) *m/z* (%): 290 (M^+(^{80}Se, ^{35}Cl) or M^+(^{78}Se, ^{37}Cl) or M^+(^{77}Se, ^{37}Cl)+1, 97.00), 288 (M^+(^{78}Se, ^{35}Cl) or M^+(^{76}Se, ^{37}Cl) or M^+(^{77}Se, ^{35}Cl)+1, 48.00), 292 (M^+(^{80}Se, ^{37}Cl) or M^+(^{82}Se, ^{35}Cl), 44.00), 286 (M^+(^{76}Se, ^{35}Cl) or M^+(^{74}Se, ^{37}Cl), 15.00), 287 (M^+(^{77}Se, ^{35}Cl) or M^+(^{76}Se, ^{35}Cl)+1 or M^+(^{74}Se, ^{37}Cl)+1, 17.00), 284 (M^+(^{74}Se, ^{35}Cl), 1.00), 291 (M^+(^{80}Se, ^{35}Cl)+1 or M^+(^{78}Se, ^{37}Cl)+1, 12.00), 289 (M^+(^{77}Se, ^{37}Cl) or M^+(^{78}Se, ^{35}Cl)+1 or M^+(^{76}Se, ^{37}Cl)+1, 11.00), 293 (M^+(^{80}Se, ^{37}Cl)+1 or M^+(^{78}Se, ^{37}Cl)+1, 6.00), 294 (M^+(^{82}Se, ^{37}Cl), 6.00), 158 (100); IR ν (cm^{-1}): 2955, 1701, 1598, 1437, 1329, 1271, 1201, 1021; HRMS calcd for $C_{11}H_{11}O_2^{80}Se^{35}ClNa^+$ (M^++Na): 312.9508. Found: 312.9523.

4.1.20. 4-Chloro-3-phenylselanylbut-2(*Z*)-en-1-ol (4t).

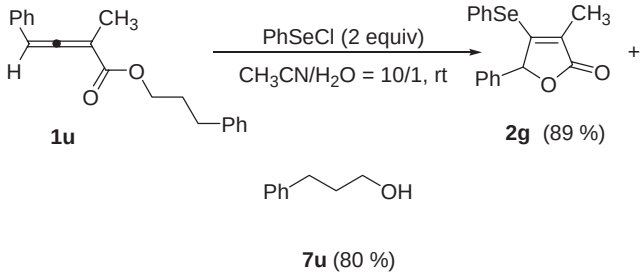


$C_{10}H_{11}ClOSe$

A solution of **3t** (103.6 mg, 0.36 mmol) in 2 mL of toluene was added 2 mL of DIBAl-H (1.0 M in toluene) at −78 °C followed by stirring at this temperature for 3 h. The mixture was quenched with 2 mL of MeOH and 5 mL of water. The mixture was then extracted with diethyl ether (25 mL×3) and dried over anhydrous Na_2SO_4. Evaporation and column chromatography on silica gel (petroleum ether/ethyl acetate=5:1) afforded **4t** (78.9 mg, 84%) as a liquid. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.40–7.37 (m, 2H), 7.22–7.18 (m, 3H), 6.42 (t, J=6.2 Hz, 1H), 4.33 (d, J=6.4 Hz, 2H), 4.05 (d, J=1.2 Hz, 2H), 2.13 (s, 1H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 137.3, 132.9, 129.9, 129.4, 128.1, 127.8, 62.1, 48.8; MS (EI) *m/z* (%): 262 (M^+(^{80}Se, ^{35}Cl) or M^+(^{78}Se, ^{37}Cl) or M^+(^{77}Se, ^{37}Cl)+1, 47.09), 260 (M^+(^{78}Se, ^{35}Cl) or M^+(^{76}Se, ^{37}Cl) or M^+(^{77}Se, ^{35}Cl)+1, 22.49), 264 (M^+(^{80}Se, ^{37}Cl) or M^+(^{82}Se, ^{35}Cl), 44.00), 258 (M^+(^{76}Se, ^{35}Cl) or M^+(^{74}Se, ^{37}Cl), 7.79), 259 (M^+(^{77}Se, ^{35}Cl) or M^+(^{76}Se, ^{35}Cl)+1 or M^+(^{74}Se, ^{37}Cl)+1, 7.19), 256 (M^+(^{74}Se, ^{35}Cl), 0.71), 263 (M^+(^{80}Se, ^{35}Cl)+1 or M^+(^{78}Se, ^{37}Cl)+1, 5.01), 261 (M^+(^{77}Se, ^{37}Cl) or M^+(^{78}Se, ^{35}Cl)+1 or M^+(^{76}Se, ^{37}Cl)+1, 4.57), 265 (M^+(^{80}Se, ^{37}Cl)+1 or M^+(^{78}Se, ^{37}Cl)+1, 2.26), 266 (M^+(^{82}Se, ^{37}Cl), 2.48), 158 (100); IR ν (cm^{-1}): 3356, 1578,

1476, 1438, 1261, 1071, 1021; HRMS calcd for $C_{10}H_{11}O^{80}Se^{35}ClNa^{+}$ ($M^{+}+Na$): 284.9561. Found: 284.9572.

4.2. Mechanistic study



The reaction of 59 mg (0.20 mmol) of **1u** and 77 mg (0.40 mmol) of PhSeCl in 0.4 mL of H_2O and 4 mL of MeCN afforded 59 mg (89%) of **2g** and 22 mg (80%) of 3-phenylpropanol **7u**. The analytic data for compound **2g** are the same as reported in Section 4.1.7. **7u** ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.27–7.19 (m, 2H), 7.16–7.11 (m, 3H), 3.62 (t, $J=6.3$ Hz, 2H), 2.65 (t, $J=7.8$ Hz, 2H), 1.89–1.81 (m, 2H), 1.50 (s, 1H); ^{13}C NMR (75 MHz, $CDCl_3$) δ 141.8, 128.39, 128.37, 125.8, 62.2, 34.2, 32.0; MS (EI) *m/z* (%): 136 (M^{+}, 13.72), 91 (100); IR ν (cm^{-1}): 3339, 2939, 1496, 1454, 1059, 1031.

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